

KATRAVATH SRINU

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EDUCATION

Indian Institute of Technology (BHU), Varanasi

Varanasi, India

B.Tech in Computer Science and Engineering

Expected May 2027

- **Coursework:** Data Structures & Algorithms, Operating Systems, DBMS, Computer Networks, AI, Machine Learning, Generative AI, Probability & Statistics
- Secured **AIR 11,227** in JEE Advanced 2023 among 150,000+ candidates

TECHNICAL SKILLS

Languages: Python, C++, JavaScript, SQL, HTML, CSS
Frameworks & Libs: Streamlit, LangChain, Scikit-learn, Pillow, Node.js, Express.js
AI / ML: LLMs, NLP, RAG, Prompt Engineering, Local Model Inference, AI Image Generation
Backend & APIs: REST APIs, WebSockets, Session Management, Modular API Client Architecture
Cloud & DevOps: AWS (EC2, Lambda, API Gateway, S3), Docker, CI/CD Pipelines, Linux
Databases: MongoDB, MySQL, Pinecone (Vector DB)
Tools: Git, Postman, Wireshark, Agile/Scrum
Core CS: Object-Oriented Design, Data Structures, Algorithm Design & Analysis, Complexity Analysis

EXPERIENCE

Independent Open Source Contributor

Remote

Backend Software Engineer — Multiple Repositories

Jan 2025 – Mar 2026

- Contributed **150+ pull requests** improving backend functionality, API performance, and code quality across open-source repositories in production; collaborated with distributed teams via **Git workflows** including feature branching, code review, and issue tracking
- Built and extended **automated test suites** to maintain reliability and prevent regressions; optimized REST API response logic to reduce latency and improve throughput across multiple service endpoints

PROJECTS

Enterprise AI Platform | *Python, Streamlit, LangChain, LLMs, REST APIs, Pillow*

2026

- Architected a **production-grade multi-module AI platform** with an LLM-powered chat assistant, AI image generation studio, analytics dashboard with natural language chart generation, and a **prompt-to-website builder** generating functional websites in under 10 minutes from a single user prompt
- Designed a **modular 5-module architecture** with session-based authentication, real-time state management, and a decoupled REST API client layer integrating LangChain, Streamlit, and Pillow for an end-to-end AI-driven user experience

VoIP Intrusion Detection System | *Python, SIP, PCAP, Wireshark, Scikit-learn*

2025

- Developed a **real-time network security pipeline** achieving **94% attack classification accuracy** across SPIT, REGISTER flooding, and SIP spoofing scenarios; analyzed **10+ GB of PCAP traffic traces** to identify adversarial patterns
- Engineered packet-level feature extraction reducing false positives by **40%**; applied algorithm optimization and complexity analysis for efficient detection on high-throughput network data streams

Real-Time Chat Application | *Node.js, Express.js, WebSockets, MongoDB*

2024

- Built a **scalable multi-user messaging platform** using WebSocket-based architecture for low-latency, concurrent real-time communication; designed with object-oriented principles and a modular service structure
- Implemented **RESTful authentication** with persistent message storage via MongoDB; architected for horizontal scalability to support enterprise-level concurrent user loads

CERTIFICATIONS

HackerRank — Software Engineer Intern Role Certification

Jan 2026

Proficiency in Data Structures, Algorithms, SQL, and Python

ACHIEVEMENTS

- Solved **250+ DSA problems** on LeetCode and Codeforces; Codeforces Max Rating: **1200**
- Secured **AIR 11,227** in JEE Advanced 2023 among 150,000+ candidates; admitted to B.Tech CSE at IIT (BHU)
- Active open-source contributor with **150+ merged pull requests** across multiple production repositories